

Thompson Sampling for Dynamic Pricing — from theory to practice - DRAFT

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Constructive review and comments are appreciated

1.Introduction

Optimal pricing strategies, such as dynamic, seasonal, or peak-and-off-peak pricing, leverage algorithmic approaches to improve profitability. Even small improvements in price can have a disproportionately large impact on profits, often exceeding the effect of comparable changes in volume or cost [1]. As of 2021, approximately 21% of companies reported using dynamic pricing [2]. Historically, modern dynamic pricing was born in the airline industry, where Smith–Leimkuhler–Darrow (American Airlines) [3] reported a \$1.4 billion gain over three years in the late 1980s. Total system revenue was \$16 to \$17 billion annually, therefore lifting total revenue by approximately 3%, corresponding to roughly \$500 million in annual incremental revenue.

A recently popular approach for sequential price optimization is Thompson Sampling (TS) [4]. Though Thompson Sampling was first introduced in 1933, its application to pricing is relatively recent [5, 6]. A more general survey tracing the origins of dynamic pricing, including historical references dating back to the 1800s through modern developments, is provided by den Boer [7]. Phillips, *Pricing and Revenue Optimization*, provides a thorough introduction to pricing theory, demand modeling, and revenue optimization, including concepts that underlie dynamic pricing [8].

Successfully applying Thompson Sampling to pricing requires careful integration of statistical modeling, Bayesian inference, algorithm design, and software implementation. Existing papers and blog posts often focus on isolated aspects of the problem, while glossing over details outside their respective domains. This paper begins from first principles, develops a Thompson Sampling-based Bayesian pricing model step by step, and integrates the resulting components into a working framework. To our knowledge, this is one of the few end-to-end treatments that rigorously connects these elements for profit-optimizing dynamic pricing. In particular, the paper is intended to provide readers, especially those not deeply familiar with all three areas, with a path from foundational principles to practical implementation.

It is important to understand prior applications of Thompson Sampling to dynamic pricing, as discussed by Ganti et al. [5] and Ferreira et al. [6], and how they relate to and differ from this work. Ganti et al. optimize revenue for baskets of goods, with examples ranging from a few items to hundreds of items, making the approach relevant to small- and large-scale e-commerce settings. In their formulation, demand variation is modeled using additive Normal variability. Ferreira et al. apply Thompson Sampling to network revenue management, revenue optimization, for a small number of products under finite inventory constraints over a selling season, such as a period of several months. Practical applications include e-commerce, hotel bookings, and other similar settings. In practice, scalability is limited to tens of products and resources rather than large-scale product catalogs. Demand uncertainty is modeled through stochastic arrivals, such as Bernoulli or Poisson demand.

In contrast to [5,6], this work applies Thompson Sampling to a single good or service to optimize profit and explicitly models stochastic demand shocks. Thus, the modeling goals of the previous works differ from the work presented here. The multiplicative variability stochastic demand model used in this work is intended to capture realistic period-to-period demand variation, where factors such as weather, seasonality, promotions, or consumer behavior can significantly change demand levels. The application of this algorithmic framework is especially useful for determining the profit-maximizing price for a good or service before introducing additional complexities such as network

constraints, as in Ferreira et al. [6], or market-basket analysis, as in Ganti et al. [5]. Furthermore, the emphasis of this work is on an integral treatment across the respective domains. Hence, in contrast to the previous works, this work focuses on the statistical demand model, posterior learning of elasticity and demand level, expected-profit optimization, and convergence behavior.

Accompanying this paper is a GitHub repository containing exemplary implementations, including:

- TS Dynamic Pricing Implementation (Jupyter Notebook) [9]
- TS Simulation Results Analysis (Jupyter Notebook) [10]

The subsequent sections address the following topics.

In Section 2, starting with the well-known constant-elasticity demand function, a stochastic demand model is developed. The formulation highlights log-normal demand with multiplicative variation and clarifies the relationships between classical demand and stochastic demand. An exemplary product or service is introduced to support examples throughout the paper. These demand expressions, together with the corresponding price-profit relationships, provide a key foundation for the Thompson Sampling pricing algorithm developed in this work.

In Section 3, beginning with a general introduction to Thompson Sampling, a TS-based sequential pricing model is developed, including the mathematical expression for the demand likelihood, prior probabilities, and corresponding model parameters. Visualizing the likelihood, priors, and model parameters relevant to our example provides insight into effective choices for the model’s density functions and parameters.

In Section 4, we describe the software implementation based on the Bayesian statistical model. The software model is set up and initialized and supports the iterative application of Thompson Sampling, including enhancements to price exploration, exploitation, model update, and optimal price estimation.

In Section 5, the iterative software model is then applied to our demand example. To gain insight into the algorithm’s convergence, we visualize the results with several graphics, including a step-by-step view of the price and profit convergence, HDI (Highest Density Interval), and KDE plots to observe the corresponding posterior density tightening.

Section 6 concludes by summarizing insights for modeling posterior demand statistics and creating an effective Bayesian pricing model. Key insights are found in the following areas: the impacts of demand shocks on the median and mean of stochastic demand, estimating input parameters, improving price exploration, estimating the optimal price, and algorithm convergence.

2. Microeconomic Demand

In microeconomics, the constant elasticity of demand function, $D(p)$ [8,11] yields the quantity demanded as a function of price, p ,

$$D(p) = ap^{-v} \tag{1}$$

Demand is modeled as the price, p , raised to the power $-v$, multiplied by the demand-scaling parameter a . The pricing algorithm developed in this paper applies to cases where $v > 1$, corresponding to price-elastic demand: a 1% increase in price results in an approximately $v\%$ decrease in quantity demanded. For convenience, economists typically write the constant-elasticity demand function with a negative sign on the exponent. Therefore, the coefficient v represents the absolute value, or non-negative magnitude, of the price elasticity.

2.1 Probabilistic Demand

In reality, demand is not deterministic. Typically, demand is modeled as a random variable, D , with multiplicative variation [12, 13]. A log-normal distribution arises naturally when variability enters a system multiplicatively, where, on a log scale, multiplicative variation becomes additive [14, 15]. In practice, demand experiences multiplicative stochastic shocks, so random disturbances scale demand up or down relative to the deterministic demand level. These unpredictable influences are traditionally called “shocks”, and in our case, “demand shocks”, arising from factors such as weather, consumer mood, competition, or news.

Mathematically, we express an observation of demand, D , as y_i , and thereby express the statistics of D as follows.

$$y_i = ap_i^{-v} e^{\varepsilon_i}, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \quad (2)$$

This defines a generative model in which demand is obtained by scaling deterministic demand with a log-normal shock. Equation 2 explicitly expresses the variation as multiplicative, representative of “demand shock”, and the exponent is zero-mean, normally distributed with standard deviation σ . These are the characteristics of a log-normal random variable.

Demand is therefore modeled as a log-normal random variable:

$$y_i | p_i \sim \text{LogNormal}(\mu_L, \sigma^2), \quad (3a)$$

where $\mu_L = \ln(\mu) = \ln(ap^{-v}) = \ln(a) - v \ln(p_i)$.

For a log-normal distribution, the median is $e^{\mu_L} = \mu = ap^{-v}$, and

$$p(y_i | a, v, \sigma, p_i) = \frac{1}{y_i \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln y_i - \mu_L)^2}{2\sigma^2}\right), \quad y_i > 0 \quad (3b)$$

These expressions represent the probability density of demand D , or equivalently, an observation y_i (a realization of D); we also use y to denote a generic observation. The conditional parameters are denoted by $\theta = (a, v, \sigma)$.

It is important to distinguish between the linear-space parameter $\mu = ap^{-v}$, Eq. (1), and its log-space counterpart $\mu_L = \ln(\mu)$. In practice, demand is observed, and decisions are made in linear space, where μ corresponds to the median demand. However, estimation and inference are performed in log-space, where the model becomes linear with additive Gaussian variations. This distinction is particularly relevant for algorithmic implementations, where parameter updates occur in log-space while predictions and decisions are evaluated in linear space.

For convenience, we express expectations for D (linear-space) or $\ln D$ (log-space), which we will refer to throughout the paper.

$$\mathbb{E}[\ln D] = \mu_L = \ln a - v \ln p \quad (4a)$$

$$\text{SD}[\ln D] = \sigma \quad (4b)$$

$$\text{Mode } D = ap^{-v} e^{-\sigma^2} = \mu e^{-\sigma^2} \text{ (peak of density)} \quad (4c)$$

$$\text{Median } D = ap^{-v} = \mu \quad (4d)$$

$$\text{Mean } D = \mathbb{E}[D] = e^{\mu_L + \frac{\sigma^2}{2}} = ap^{-v} e^{\frac{\sigma^2}{2}} = \mu e^{\frac{\sigma^2}{2}} \quad (4e)$$

$$\text{SD}[D] = \mu e^{\sigma^2/2} \sqrt{e^{\sigma^2} - 1} \quad (4f)$$

$$\text{Demand Coefficient of Variation, } CV_d = \frac{\text{SD}[D]}{\mathbb{E}[D]} = \sqrt{e^{\sigma^2} - 1} \quad (4g)$$

We observe that the log-normal density $p(D)$ is skewed to the right and that $\text{Mode} < \text{Median} < \text{Mean}$. Due to log-normality, the expected demand exceeds the median: $\mathbb{E}[D] = \mu e^{\sigma^2/2} > \mu$. That is, the mean is shifted to the right by the multiplicative factor and the median aligns with the deterministic demand, Eq. (1). This distinction is important for pricing decisions, where expected revenue depends on $\mathbb{E}[D]$, not the median.

Regression models of demand typically transform demand to log-scale, with the transform $\ln D = \ln(D)$, where $\ln D$ is normally distributed with an additive variation. Log-log regression is applied, and predictions in log-space, $\ln \hat{D}$,

are transformed back to a linear scale with the exponential transform $e^{\ln \hat{D}}$, which corresponds to the median in linear space, while the expected (mean) demand is $e^{\hat{\mu}_L + \sigma^2/2}$ [16, 17].

The log-space Demand is given by

$$\ln y \sim \mathcal{N}(\mu_L, \sigma^2) \quad (5)$$

and the Normal density function corresponding to the log demand is

$$p(\ln y \mid \theta, p) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu_L)^2}{2\sigma^2}\right) \quad (6)$$

Note that the log-normal density Eq. (3) and the normal density Eq. (6) are both functions of $\ln y$. This dependency arises because, typically, a log-normal random variable is formulated in log-space, as in Eq. (5). Transformation to linear-space $D = e^{\ln D}$, is a change of variable resulting in the term $\ln y$ in the exponent, and the factor $1/y$ in the denominator, Eq. (3b).

2.2 E-commerce/Retail Demand Example

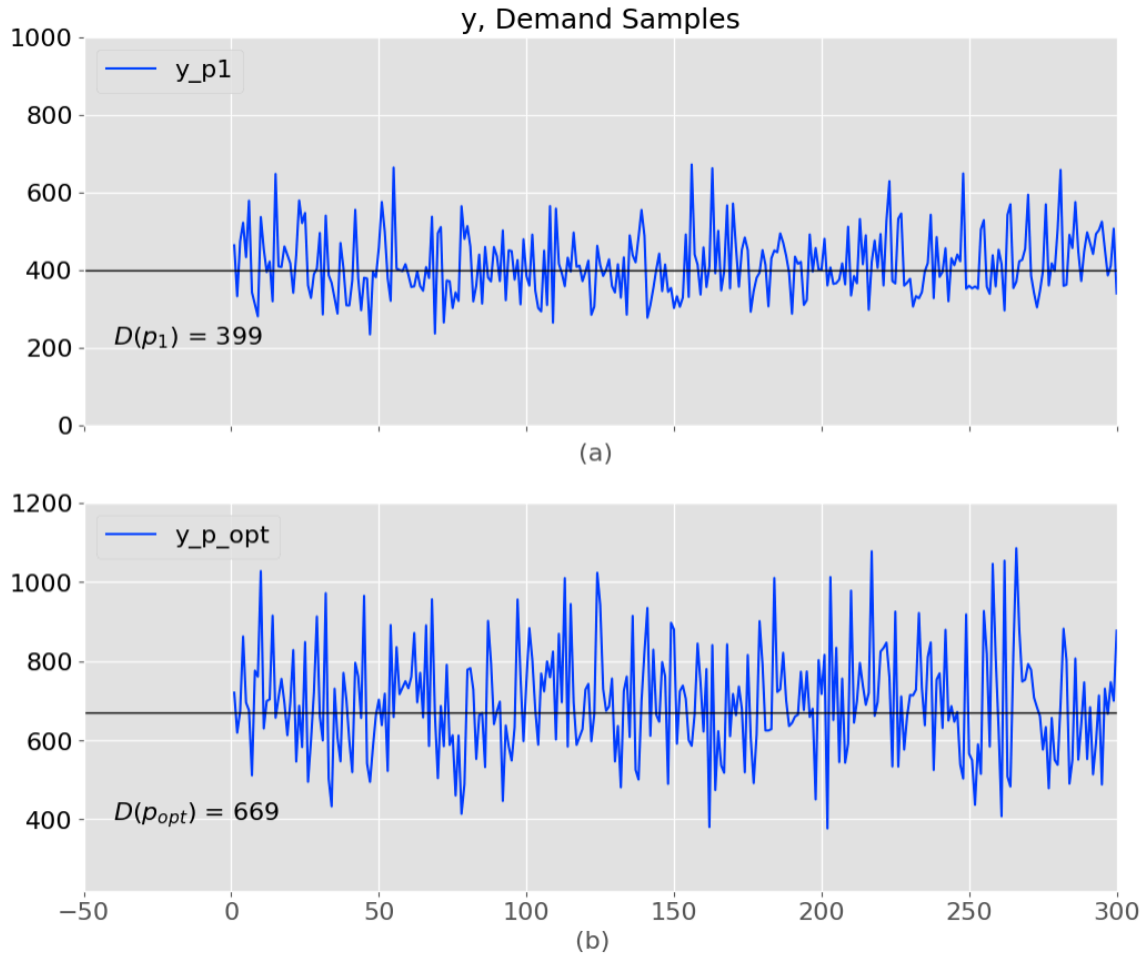


Figure 1: Each plot shows demand samples over 300 periods ($a = 4000$, $v = 1.8$, $CV_d = 0.2$) at two price levels: (a) $p_1 = \$3.6$ and (b) $p_{opt} = \$2.7$.

As a running example throughout this paper, we assume a demand elasticity parameter of $v = 1.8$, consistent with empirical estimates for discretionary e-commerce categories and digital goods. For such goods, observed price

elasticities typically range from -1.2 to -1.7 (with v denoting the absolute value), with higher sensitivities in highly substitutable and promotion-driven markets [18–22].

Demand example parameters:

$$\begin{aligned} a &= 4000, \text{ demand shaping parameter} \\ v &= 1.8, \text{ demand elasticity} \\ CV_d &= 0.2 \end{aligned}$$

We further assume a coefficient of variation $CV_d = 0.2$, corresponding to a moderate and fairly stable demand (i.e., a 20% standard deviation relative to mean demand), which is common for mature retail categories and recurring services.

The demand at two price levels is illustrated in Figure 1, a and b, contrasting the initial offered price, $p_1 = \$3.6$ vs the optimum price $\$2.7$ (see next section). On each graph, the deterministic demand (ap^{-v}) is indicated by the horizontal black line, which corresponds to the median of the demand distribution. In Figure 1a, the mean demand at the price $\$3.6$ is 399, which is significantly below $D_{\text{opt}} = 669$ at p_{opt} (Fig. 1b). At p_{opt} , the variability increases and the distribution is right-skewed, with the mean exceeding D_{opt} due to the log-normal factor, Eq. (4e).

2.3 Demand, Price, and Profit

The objective of the dynamic pricing algorithm is to find the price that optimizes profit. Therefore, it is necessary to understand the precise relationship between price and profit. The derivation of the price for optimal profit can be found in microeconomic textbooks, such as [11]. Here, we assume that the cost of production is governed by fixed cost, F (\$6), and variable unit cost, c (cost per unit, \$1.2). Furthermore, we assume a monopolistic pricing situation, the typical assumption for this analysis. That is, competitors or substitute goods do not affect demand, and thus the firm offering the product can follow its own demand curve, unperturbed by competitors' responses to its price variations. The TS Algorithm is resilient to inaccuracies in this assumption, as the data gathered at each price point includes competitor responses and potential product substitutions.

We express the profit as the revenue minus the cost

$$\text{Profit} = \text{Revenue} - \text{Variable Costs} - \text{Fixed Costs}$$

Under stochastic demand, the median and mean revenue and costs are

$$\begin{aligned} \text{Revenue (median)} &= p \cdot D(p) \\ \text{Revenue (mean)} &= p \cdot D(p)e^{\sigma^2/2} \\ \text{Variable Costs (median)} &= c \cdot D(p) \\ \text{Variable Costs (mean)} &= c \cdot D(p)e^{\sigma^2/2} \\ \text{Fixed Costs} &= F \end{aligned}$$

The median profit is

$$\text{Median Profit}(p) = (p - c)D(p) - F \quad (7a)$$

This is equivalent to the classic profit equation (deterministic demand, Eq. (1)). However, we see that the expected profit is greater than the median due to the multiplicative log-normal factor [8, 14, 15], $e^{\sigma^2/2}$

$$\text{Mean Profit}(p) = (p - c)D(p)e^{\sigma^2/2} - F \quad (7b)$$

Under log-normal demand, multiplicative demand variation results in greater expected demand due to the distribution's right skewness; correspondingly, expected revenue and profit are greater than their respective medians. The multiplicative demand shocks, e^{ϵ_i} (Eq. (2)), are strictly positive and bounded below by zero, but unbounded above. This asymmetry raises the mean above the median.

The log-normal mean differs from the median by a multiplicative constant, but the multiplicative term is independent of price. Hence the profit-maximizing price is identical under both formulations. Starting with Eq. 7a or 7b, differentiating with respect to price and solving for the optimal price, we obtain

$$p^* = \frac{v}{v-1}c \quad (8)$$

For our running example, with the help of Eqs. (1), (7a), (7b), and (8), we calculate the optimum price, demand, and profit:

$$Price_{\text{opt}} = p^* \approx \$2.7$$

The median demand and profit are

$$D_{\text{opt}} \text{ median} \approx 669$$

$$Profit_{\text{opt}} \text{ median} \approx \$997$$

The mean demand and profit are

$$D_{\text{opt}} \text{ mean} \approx 683$$

$$Profit_{\text{opt}} \text{ mean} \approx \$1018$$

Figure 2 presents classic illustrations of demand (median), revenue, costs, and profit. The initial offered price is $p_1 = \$3.6$, and the profit is represented by the area of the shaded region (Figure 2a), $\$399 \times (3.6 - 1.22) \approx \950 . Here, $\$1.22$ represents the average cost per unit, which includes both variable cost and fixed cost amortized over demand, i.e., $c + F/D$.

At the optimal price $p_{\text{opt}} = \$2.7$ (Eq. 8), the higher demand reduces the fixed cost per unit (i.e., F/D), resulting in an average cost of approximately $\$1.21$. The corresponding profit is $669 \times (2.7 - 1.21) \approx \997 , Eq. (7).



Figure 2: (a) Median demand vs. price, showing revenue, cost, and profit at $p_1 = \$3.6$ and (b) $p_{\text{opt}} = \$2.7$.

Thus, the objective of the TS Dynamic Pricing algorithm is to learn this optimal price through sequential observations of demand, that is, to move from the initial price toward the optimum price. In our running example, this corresponds to median profit \$950 at the initial offered price of $p_1 = \$3.6$ to a median profit of \$997 (mean profit \$1018) at the offered price of $p_{\text{opt}} = \$2.7$.

3. Thompson Sampling

3.1 Introduction to Thompson Sampling

Before describing TS applied to dynamic pricing, we briefly review Thompson Sampling and some use cases where it is applied.

The TS method solves the so-called “multi-armed bandit” (MAB) problem. This terminology originates from a scenario [23] in which a gambler sits down at a slot machine with multiple levers (i.e., arms). When pulled, an arm produces a random payout. Because the distribution (i.e., probability) of payouts corresponding to each arm is unknown, the player can learn this distribution only by experimenting. As the gambler learns about the arms’ payouts, she faces a dilemma: in the immediate future, she expects to earn more by exploiting arms that have yielded high historical payouts, but by continuing to explore alternative arms, she may learn how to earn higher payouts in the future.

For the ensuing discussion, it is helpful to recall Bayes theorem and the corresponding terminology.

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} = \frac{\text{Likelihood} \cdot \text{Prior}}{\text{Evidence}} \quad (9)$$

TS is a Bayesian algorithm; that is, it applies Bayes' rule to update the posterior probability distribution, which is decomposed (by Bayes' rule) into the corresponding likelihood and prior probabilities. On the right side of the expression, the posterior distribution is expressed in terms of likelihood, prior, and evidence.

Typical use cases in which Thompson Sampling provides a solution include A/B testing and Recommendation Systems.

- TS enables dynamic traffic allocation to the different groups based on Bayesian probabilities. For example, the audience is divided into two groups (A and B), each exposed to different variants of the product or website
- In recommendation systems, TS is applied to explore and exploit user preferences efficiently. These systems aim to present personalized content or items to users based on their historical interactions or preferences.

TS implements the exploration–exploitation tradeoff using a Bayesian approach. At each step, it samples model parameters from the posterior distribution and selects the action that is optimal under the sampled parameters (“exploitation”). To learn the underlying statistics, TS explores the parameter space through posterior sampling while simultaneously exploiting current estimates. As the posterior distribution tightens, samples concentrate around the true parameter values, leading to more consistent selection of optimal actions.

When applied to dynamic pricing, an attractive feature of the TS algorithm is its price exploration, which provides a systematic approach to pricing.

3.2 TS Pricing Model

3.2.1 Bayesian Stochastic Demand

We formulate our Bayesian model based on the demand posterior, $p(\theta | \mathbf{D})$. The demand, $\mathbf{D}$ is written in bold font because within the context of our TS algorithm, it represents a set of demands, $\mathbf{D} = (y_1, \dots, y_n)$, that is, a demand measurement, y_i , at each step of the iterative process. Furthermore, the TS algorithm is formulated to discover the demand parameters a and v in the deterministic demand model, Eq. (1). Applying Bayes rule, and accounting for the iterative price and demand observations, we obtain an expression for the posterior distribution.

$$p(\theta | \mathbf{D}) \propto \left(\prod_i^n p(y_i | \theta, p_i) \right) p(a) p(v), \quad \mathbf{D} = (y_1, \dots, y_n) \quad (10)$$

where $p(y_i | \theta, p_i)$ is the log-normal likelihood defined in Eq. (3b), and observations are assumed conditionally independent. In the general demand model of Section 2.1, θ denotes the full parameter set (a, v, σ) . In our TS implementation, σ is pre-estimated and treated as a fixed input parameter; therefore, for the posterior in Eq. (10), θ denotes the learned parameters (a, v) . Thus, the posterior is proportional to the product of likelihood terms multiplied by the prior distributions $p(a)$ and $p(v)$. The price p_i at each step is the offered price, selected by price exploration.

A closed-form expression for the posterior is not analytically tractable, therefore, equation 10 is implemented with the help of a probabilistic programming language (PPL) supporting Bayesian statistical inference and optimization, such as Stan [24] or PyMC [25].

The TS dynamic pricing model iteration, each step i , includes the following actions and corresponding algorithm behavior.

- Exploration
 - At each step, TS samples parameters $\theta = (a, v)$ from the posterior distribution and selects the price that maximizes the profit function, Eq. (7) according to the sampled parameters.
 - Our implementation introduces additional exploration heuristics to improve convergence (Section 4.3.1).
 - As the posterior tightens, samples increasingly concentrate around high-probability parameter values.
- Exploitation

- At each step, the selected price is offered to the market, and the market response (demand) is measured.
- In practice, the observed demand is the measured market response. However, in our TS iteration example (Section 5), the observed demand is simulated by sampling from a generative model defined by Eq. (2).
- Update posterior
 - Updating the posterior corresponds to including the new observation data, (p_i, y_i) in Eq. (10), and thereby refining the parameter estimates to better fit the observed data.
 - As the posterior distribution tightens, the density function (Figure 3) will shift towards the optimal demand and with increasing concentration around the optimal parameter values.

This iterative process balances exploration and exploitation, enabling convergence to the price that maximizes expected profit.

3.2.2 TS Model Input Parameters

Before a detailed discussion of each term in Eq. (10), it is useful to summarize the model’s input parameters. For convenience, all input parameters and their relationships to the model variables (Eqs. (10)–(13)) are listed in Table 1. The details of the corresponding prior probability density functions and parameters are discussed in the following subsections. These priors encode initial beliefs about the demand parameters before applying the pricing model. The parameter values are chosen to align with the demand example in Section 2.2.

Table 1: Model input parameters and values

Model Variable	Prior Distribution / Function	Input Parameters	Description
a	Log-normal	$\mu_{a,\text{linear}} = 3900,$ $CV_a = 0.05$	Shaping parameter a is modeled with a log-normal prior, parameterized by its mean in linear space and coefficient of variation.
v	Truncated normal	$m_v = 1.5, \sigma_v = 0.4,$ lower bound = 1.1	Demand elasticity v is modeled with a truncated normal distribution with mean m_v , standard deviation σ_v , and lower bound 1.1.
σ	Fixed	$CV_d = 0.18$	Demand variance σ is treated as a fixed parameter, specified via the coefficient of variation of demand, CV_d .
p_i	$[p_{\min}, p_{\max}]$	$[\$2, \$4]$	The minimum and maximum prices define the allowable range for price exploration.

3.2.3 Demand Likelihood

The demand likelihood is the product of log-normal probabilities corresponding to all data points $\{(p_i, y_i)\}_{i=1}^n$, where n corresponds to the current step.

$$p(y_i | \theta, p_i) = \prod_{i=1}^n \frac{1}{y_i \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln y_i - \ln \mu_i)^2}{2\sigma^2}\right), \quad \mu_i = ap_i^{-v} \quad (11a)$$

The likelihood for $i = 1$ is graphed in Figure 3, where $p_1 = \$3.6$. We observe that the likelihood is right-skewed, Eq. (4e), so the peak corresponds to the mode, $D_{\text{mode}} | p_1 = 383$, Eq. (4c). Also shown on the graph is the optimum (median) demand, $D_{\text{opt}} = 669$, at price $p_{\text{opt}} = \$2.7$. The figure illustrates the initial likelihood corresponding to the first observation. As additional data are observed, the posterior distribution over the parameters shifts and concentrates, leading to improved estimates of the optimal demand D_{opt} .

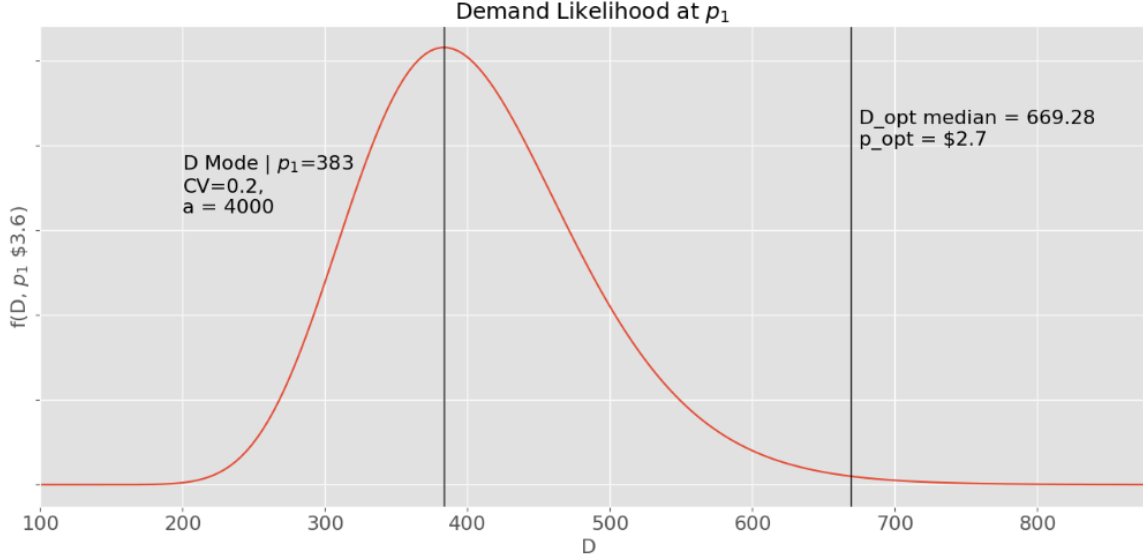


Figure 3: Demand likelihood at price $p_1 = \$3.6$, with $a_{\text{true}} = 4000$ and $CV_d = 0.18$.

The parameters corresponding to our example applied to Eq. (11) are listed in Table 1. The demand variation σ is derived from the coefficient of variation input $CV_d = 0.18$, according to Eq. (4g). In contrast, the true coefficient of variation is $CV_{d,\text{true}} = 0.2$. Thus, $CV_{d,\text{true}} = 0.2$ is employed in the simulated demand data generation (TS iteration example, below). However, since $CV_{d,\text{true}}$ is unknown to the algorithm (i.e., pricing model), the model uses a fixed estimate $CV_d = 0.18$ as an input parameter.

3.2.4 Elasticity of Demand, Prior v

The probability density $p(v)$ represents our understanding of the elasticity before applying the pricing model. The elasticity of demand in our example case is elastic ($v > 1$) so that a 1% increase in price results in an approximately $v\%$ decrease in quantity demanded. Furthermore, the optimal price formula, Eq. (8), has a singularity at $v = 1$ (unit elasticity). Gelman et al. (Bayesian Data Analysis [26]) recommends constrained truncated priors when parameters are only meaningful on restricted domains and when unconstrained priors induce pathological behavior. Therefore, we restrict v by modeling the prior, $p(v)$, as a truncated normal distribution, with a lower limit of $v_{\min} = 1.1$. This constraint ensures economically meaningful elasticity values and avoids unstable pricing behavior near $v = 1$.

The truncated-normal density is given by

$$p(v) = \frac{1}{C \sigma_v \sqrt{2\pi}} \exp\left(-\frac{(v - \mu_v)^2}{2\sigma_v^2}\right), \quad v > v_{\min} \quad (12a)$$

where C is the normalizing constant, equal to the survival function of a standard normal distribution:

$$C = 1 - \Phi\left(\frac{v_{\min} - \mu_v}{\sigma_v}\right) \quad (12b)$$

Φ is the CDF (Cumulative Distribution Function) of the standard normal distribution.

Figure 4 illustrates the truncated normal prior density $p(v)$, with parameters corresponding to our example case (Table 1). The true elasticity $v = 1.8$ is not known to the algorithm (i.e., pricing model). Therefore, before applying the algorithm, an initial prior over v is specified, and is comprised of parameters $\mu_v = 1.5$ and $\sigma_v = 0.4$, with lower limit $v_{\min} = 1.1$.

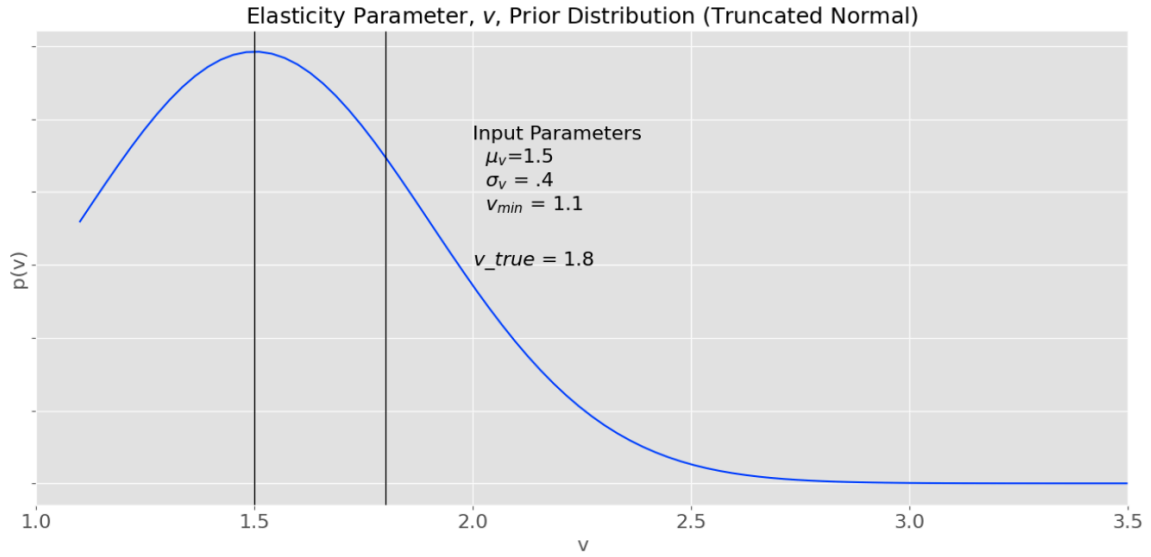


Figure 4: Prior distribution of elasticity v (truncated normal) with input parameters.

3.2.5 Shaping Parameter Prior, a

The probability density $p(a)$ represents our understanding of the demand multiplicative constant a before applying the pricing model. The constant a is strictly positive, and therefore the recommended prior model is log-normal [27].

$$p(a) = \frac{1}{a\sqrt{2\pi}\sigma_a} \exp\left(-\frac{(\ln a - \mu_a)^2}{2\sigma_a^2}\right) \quad (13a)$$

$$a \sim \text{LogNormal}(\mu_a, \sigma_a^2) \quad (13b)$$

$$\ln a \sim \mathcal{N}(\mu_a, \sigma_a^2) \quad (13c)$$

Equations (13b-13c) clarify that the log-normal prior is parameterized in log space.

Before applying the model, an initial prior over a is specified based on parameter estimation, with parameters $\mu_{a,\text{linear}} = 3900$ and $CV_a = 0.05$.

Accurate convergence in the dynamic pricing model depends on a well-specified initial prior for a . In particular, priors centered near the true value, $\mu_{a,\text{linear}}$, with moderate variance, moderate CV_a , lead to improved posterior parameter estimation. It is observed empirically that a wider initial prior over a (larger CV_a) increases uncertainty in the demand estimate, which can propagate to the estimation of v , leading to less accurate estimates of the optimal price.

The prior probability density $p(a)$ and input parameters are visualized in Figure 5.

By convention, the log-normal parameters μ_a and σ_a denote the mean and standard deviation in log space. The linear-space mean is given by $\mathbb{E}[a] = \mu_{a,\text{linear}}$, and the median of the distribution is e^{μ_a} . The parameters $\mu_{a,\text{linear}}$ and $\sigma_{a,\text{linear}}$ are estimated from demand observations prior to applying the pricing algorithm. The corresponding log-space parameters are obtained via the following transformations:

$$\mathbb{E}[a] = \mu_{a,\text{linear}} = e^{\mu_a + \sigma_a^2/2} \quad (14a)$$

$$\sigma_a = \sqrt{\ln(1 + CV_a^2)} \quad (14b)$$

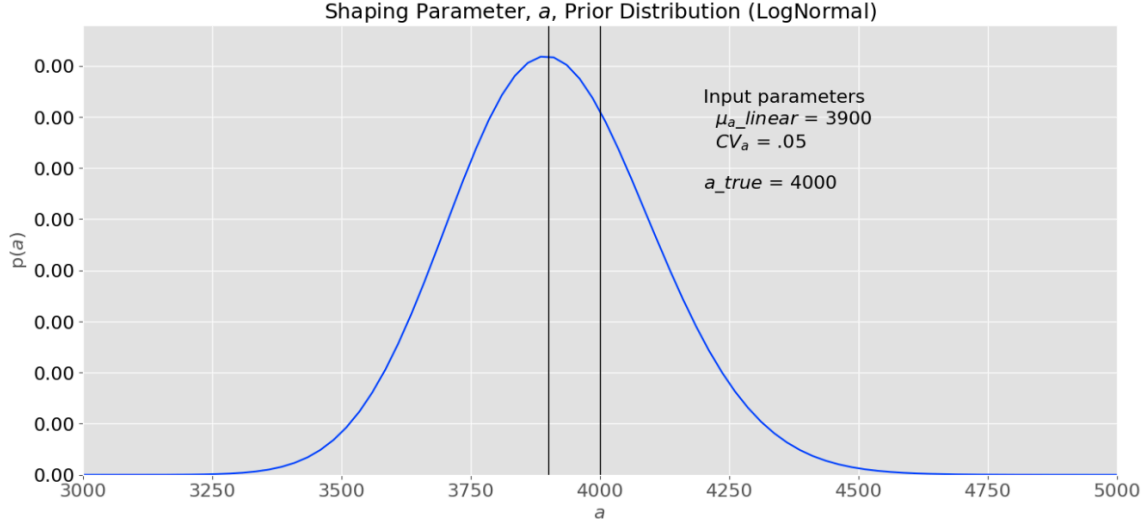


Figure 5: Shaping parameter a , log-normal prior distribution and input parameters.

$$CV_a = \frac{\sigma_{a,\text{linear}}}{\mu_{a,\text{linear}}} \quad (14c)$$

$$\mu_a = \ln(\mu_{a,\text{linear}}) - \sigma_a^2/2 \quad (14d)$$

The subtraction term in Eq. (14d), $-\sigma_a^2/2$ arises because the term $\sigma_a^2/2$ must be subtracted from linear space mean $\mathbb{E}[a] = e^{\mu_a + \sigma_a^2/2}$ to obtain the log space mean μ_a . In contrast, in Eq. (3b), $\mu_L = \ln(\mu)$ corresponds to the log of the median (deterministic demand) rather than the mean, and therefore no such adjustment is required.

3.2.6 Demand Spread, σ

Demand variability is controlled by the parameter σ , of the log-normal demand model. Before applying the pricing model, CV_d is estimated from demand observations, and σ is derived according to Eq. (4g). In our example, the estimated value is $CV_d = 0.18$ (Table 1).

It is important to note that σ could be included in Eq. (10) as an additional parameter with an associated prior. However, in our experiments, doing so increases uncertainty in the demand level, leading to less accurate estimates of the optimal price. Thus, it was found more effective to estimate CV_d and compute σ accordingly.

A natural question is why the parameter a is not fixed by estimating it prior to applying the pricing algorithm, as is done here with σ . Fixing the demand-scaling parameter a would introduce systematic bias into the model. Because a directly determines the demand level, an incorrect fixed value shifts the expected demand and forces compensation through the elasticity parameter v , leading to biased estimates of v and the optimal price.

By comparison, σ governs the demand spread and does not introduce systematic bias in the mean demand. As a result, fixing σ primarily affects statistical uncertainty rather than inducing systematic error, with a comparatively smaller impact on the optimal price estimate.

4. TS Pricing Iterative Software Model

We now describe the software implementation of the iterative pricing model, including:

1. Model setup and initialization
2. Estimation of the global optimal price
3. Exploration, exploitation, and posterior update

4.1 Step 1: Model Setup and Initialization

In this step, the statistical pricing model, Eqs. (10)–(13) is set up with input parameters and initialized with the first data point (p_1, y_1) . The software model is implemented using the PyMC probabilistic programming framework [25] within a Jupyter Notebook [9]. In this work, PyMC is selected to enable seamless integration with a Python-based application stack, particularly for backend API deployment.

To illustrate the PyMC implementation, we present below (Listing 1) the Python initialization method `init_ts_pricing_model`, an excerpt from the Jupyter Notebook [9], which directly encodes the probabilistic model defined in Section 3.2.

The goal of this discussion is to provide a conceptual overview of the software and how it implements the statistical model rather than a line-by-line detailed coding explanation. The function resides within a Python class (hence the reference to `self`) and supports several options for experimentation; here, we focus on the default configuration used in this work.

Listing 1: Bayesian software model initialization.

```

1 def init_ts_pricing_model(self, draws: int = 1000, tune: int = 2000, chains: int = 2, target_accept: float
  = 0.99):
2
3     with pm.Model() as model:
4         # ---- Prior for a (LogNormal via loga ~ Normal) ----
5         cv2_a = (self.sigma_a_linear / self.m_a_linear) ** 2
6         sigma_loga = np.sqrt(np.log1p(cv2_a))
7         mu_loga = np.log(self.m_a_linear) - 0.5 * sigma_loga**2
8
9         # ---- Prior for v (truncated normal) ----
10        v = pm.TruncatedNormal("v", mu=self.m_v, sigma=self.sigma_v, lower=self.lower_v)
11
12        # ---- a (learned or fixed) ----
13        if self.m_a_linear_fixed is None:
14            a = pm.LogNormal("a", mu=mu_loga, sigma=sigma_loga)
15        else:
16            a = pm.Deterministic("a", 0 * v + self.m_a_linear_fixed)
17        loga = pm.Deterministic("loga", pm.math.log(a))
18
19        # ---- sigma_log (log-space variation, learned or fixed) ----
20        if self.sigma_log_fixed is None:
21            sigma_log_scale = np.sqrt(np.log1p(self.CV_d**2))
22            sigma_log = pm.HalfNormal("sigma_log", sigma=sigma_log_scale)
23        else:
24            sigma_log = pm.Deterministic("sigma_log", 0 * v + self.sigma_log_fixed)
25        cv = pm.Deterministic("cv", pm.math.sqrt(pm.math.exp(sigma_log**2) - 1))
26
27        self.p_data = pm.Data("p_data", np.array([self.p0], dtype=np.float64))
28        self.y_data = pm.Data("y_data", np.array([self.y0], dtype=np.float64))
29
30        mu_log = loga - v * pm.math.log(self.p_data)
31        pm.LogNormal("D_obs", mu=mu_log, sigma=sigma_log, observed=self.y_data)
32
33        self.model = model
34        self.trace = pm.sample(
35            draws=draws, tune=tune, chains=chains,
36            target_accept=target_accept,
37            random_seed=self.random_seed,
38            progressbar=self.verbose,
39        )
40
41    return self.trace

```

- Line 10 defines the elasticity parameter v using a truncated normal prior, corresponding to Eq. (12) (Figure 4).

- Line 14 defines the prior for a as a log-normal distribution, corresponding to Eq. (13) (Figure 5). For evaluation purposes, the model also supports setting a to a constant value (lines 13–16), which is useful for experimentation; however, as discussed previously, the prior formulation is used in this work.
- Lines 21–22 define the log space demand variation parameter σ based on the input CV_d . In the TS iteration, we use a fixed CV_d input. For experimentation, lines 23–24 also allow σ to be learned via a prior.
- Lines 27–28 define mutable data containers for price and demand observations.
- Line 31 specifies the demand likelihood as a log-normal distribution, with inputs including the log-scale mean, standard deviation, and observed price and demand data. This corresponds directly to Eq. (3b).
- After defining the model, previous lines, line 34 runs posterior sampling using PyMC. With the default parameters (line 1), the sampler performs a tuning phase of 2,000 warm-up steps, followed by 1,000 posterior samples per chain across two chains, yielding 2,000 posterior draws (1,000 per chain). During tuning, PyMC adapts internal sampling parameters (e.g., step size and mass matrix) to improve sampling efficiency. The resulting draws approximate the posterior distribution defined by the likelihood, priors, and observed data [23, 24].

4.2 Optimum Price Estimate

At each step, starting with step 1, the global optimum price $p_{\text{optest},i}$ is estimated. This estimate is similar, but differs from the offered price calculation. The offered price is based on a limited number of samples (draws) from the posterior distribution and applies local price constraints. The optimum price estimate is the price that maximizes expected profit obtained by averaging over the entire set of posterior draws. In mathematical terms, $p_{\text{optest},i}$ is calculated as follows.

Create a price grid

$$P_g = \{p_j\}_{j=1}^{N_p}, \quad p_j \in [p_{\min}, p_{\max}] \quad (15a)$$

For example, $N_p = 200$, where $p_{\min}$ and $p_{\max}$ are listed in Table 1.

Select the set of S posterior draws

$$\{a^{(s)}, v^{(s)}\}_{s=1}^S \quad (15b)$$

S is the number of draws, where in our case $S = 2000$ (2 chains, 1000 draws per chain). Calculate the expected profit for each price p_j using median demand (Eq. (7a)):

$$\text{Median } \mathbb{E}[\text{Profit}(p_j)] = \frac{1}{S} \sum_{s=1}^S [(p_j - c) a^{(s)} p_j^{-v^{(s)}} - F] \quad (15c)$$

Since mean demand differs from median demand by a multiplicative constant independent of price (see Section 2.3), the optimal price is unchanged whether based on mean or median demand.

The optimum price point corresponds to the maximum expected profit:

$$p_{\text{optest},i} = \arg \max_{p_j \in P_g} \mathbb{E}[\text{Profit}(p_j)] \quad (15d)$$

4.3 Steps 2 ... N

As the algorithm advances, each iteration samples parameters from the posterior, calculates a new offered price, offers the product or service to the market, observes the resulting demand response, and updates the posterior model with the new data. Using the updated posterior, the algorithm then generates a revised estimate of the optimal price. Below, we discuss each step in more detail.

4.3.1 Exploration

Exploration is especially important because it generates new offered prices, and together with exploitation, the model learns the demand statistics. For successful convergence, exploration needs to consistently explore optimal price areas. Below, we discuss several enhancements to the exploration that encourage the model to explore price areas consistent with the optimal price.

Global price limits All offered prices are restricted, clipped, to the global region $[p_{min}, p_{max}]$. This defines the global exploration range, effectively setting guardrails on the overall set of offered prices. In our example, we set the global price range to $p_{min} = \$2.00$ on the low end and $p_{max} = \$4.00$ on the high end.

Fixed Initial Prices It is helpful to seed the initial exploration with fixed price points. This is to ensure the algorithm sees the range of acceptable prices before the posterior begins to tighten, potentially missing price regions corresponding to the optimum price.

K Posterior Samples TS sampling, by default, entails sampling the posterior parameters with one sample, $K=1$. For our pricing use case, the process of computing a new offered price, begins by sampling the posterior parameters, $(a^{(k)}, v^{(k)})_{k=1}^K$, $K \geq 1$, followed by computation of the optimum profit. $K > 1$ posterior samples reduces variance in the price estimate by averaging over multiple parameter draws, thereby dampening exploration and stabilizing price updates.

For the purpose of avoiding negative customer perception, keeping the price changes constrained is desirable. Dampening the price update can be useful during early iterations, when the posterior probability is wide, as is illustrated in Figures 7 and 8. We have found that this approach, following initial fixed prices, is effective in the early algorithm steps.

Calculating the new offered price, based on K samples, is similar to estimating the global optimum price (discussed above), however the calculation is based on K posterior samples and in addition applying a local price constraint. The offered price, at step i , is calculated as follows. Randomly select K draws, and then find the price that optimizes profit.

Create a local price grid

$$P_{lg} = \{p_j\}_{j=1}^L, \quad p_j \in [p_{\text{local min}}, p_{\text{local max}}] \quad (16a)$$

for example $L = 200$. The local price deviation minimum and maximum bounds are input parameters, for example, local price deviation may be +/- 10% from the previous price.

Randomly select K posterior samples (draws),

$$\{(a^{(k)}, v^{(k)})\}_{k=1}^K \quad (16b)$$

Compute the expected profit at each grid point p_j , based on the median profit formulation (Eq. (7a)).

$$\text{Median } \mathbb{E}[Profit_K(p_j)] = \frac{1}{K} \sum_{k=1}^K [(p_j - c) a^{(k)} p_j^{-v^{(k)}} - F] \quad (16c)$$

The new offered price, p_i^* , is the price that maximizes expected profit:

$$p_i^* = \arg \max_{p_j \in P_{lg}} \mathbb{E}[Profit_K(p_j)] \quad (16d)$$

Local Price Anchor Observing the price posterior probability, as displayed in Figure 7 (below), illustrates that the KDE distribution is right-skewed. Particularly after some number of iterations, when calculated offered price from posterior parameter samples, posterior sampling can lead to an upward drift in the offered prices and thereby cause the posterior probability to drift away from the optimum price. This behavior arises due to the multiplicative stochastic demand variation and highlights the importance of understanding the effect of log-normal demand shocks on posterior price exploration.

The remedy is to anchor exploration around the current optimal price estimate. Thus, this mode of exploration does not rely on new posterior samples, but instead perturbs the current optimal price estimate +/- some deviation, alternating up + and down -.

N Price Repeat As the demand uncertainty increases, for example $CV > 0.1$, experimentation shows that it is helpful to keep the price fixed at the offered price N_price_repeat times. For example, if $N_price_repeat = 3$, the price is offered 3 times, and on the 4th iteration, a new price is offered, according to the exploration strategy - fixed initial prices, K -th posterior sample, or local price anchor. The objective is to obtain a clearer demand signal despite the demand variance. For example if $CV = 0.2$, then the $CV_{effective} \approx CV/\sqrt{N_price_repeat}$, under the assumption of independent demand realizations.

4.3.2 Exploitation

Now that a new offered price is available, it is exploited; that is, the product is offered to the market at that price. In a real-life situation, the demand is measured. In our case, we simulate the market response by sampling from a demand probability function defined by Eq. (2), according to the true value of $CV_{d,true} = 0.2$.

4.3.3 Posterior Update

The posterior update is similar to the initialization described in Section 4.1. At each step i , the posterior is updated to include the additional price and demand data, followed by 2000 tuning (warm-up) steps on 2 independent chains, and finally 2000 draws (1000 per chain).

4.3.4 Global Optimum Price Estimate

At each step, i , the global optimum price is estimated, $p_{optest,i}$ according to Eq. (15).

5. TS Iteration Results

Effective application of Thompson Sampling to dynamic pricing requires monitoring the algorithm's operation and convergence. To this end, several visual diagnostic tools are applied in order to understand the algorithm's convergence characteristics.

5.1 Price and Profit

Figure 6 shows the price at each iteration and illustrates the algorithm's ability to discover the profit optimizing price.

The model starts with the offered price $p_1 = \$3.6$. After the first posterior update, the model forms an estimate of the demand parameters. In this case, the first optimal price is estimated at $\sim \$2.64$, which is a reasonably good estimate after only one posterior update. However, each run of the algorithm will differ due to the statistical nature of the demand and the numerical posterior estimate. Across experiments, convergence is typically achieved within a few percent of the optimal price. As observed in the figure, after the initial step, the optimal price estimate oscillates around the true optimal price (p_{opt}) and reaches a very close estimate by step 15, $p_{optest} = \$2.69$. After step 15, the posterior continues to tighten, and the optimal price estimate stabilizes near the optimum.

Similarly, the profit corresponding to $p_1 = \$3.6$ is \$950 and, by iteration $i = 15$, reaches the optimum median profit, \$997 ($Profit_{opt}$), after which the posterior continues to tighten and the profit remains close to its maximum value.

The graph also illustrates the application of several enhancements to exploration, including

- *Fixed Initial Prices* - the initial prices are fixed to cover the global price range, in this case $[\$3.6, \$2.3]$.
- *Local Price Min and Max Deviations* - after the fixed initial price offers, and prior to step 15, the maximum local allowed price deviation is 100% from the previous offered price (subject to the global min and max prices). However, we observe in the figure, after the initial prices, the price changes are fairly well under control.
- *N Price Repeat* - in this case $N_price_repeat = 3$, so that the offered price is offered 3 times before a new offer.
- *K Posterior Samples* - after the initial price exploration phase (fixed initial prices), the offered price at each step (steps 7 to 14) is based on K Posterior Samples ($K=2$) and the price is estimated according to Eq. (16d).
- *Local Price Anchor* - From iteration $i = 15$, the exploration strategy switches to Local Price Anchor mode, where price updates are constrained to $\pm 5\%$ around the current optimal price estimate.

This behavior illustrates the balance between exploration and exploitation: early iterations explore the price space broadly, while later iterations refine the estimate near the optimal price.



Figure 6: TS Simulation step by step (a) offered price and price optimum estimate, (b) median profit.

5.2 Posterior Tightening

The profit-maximizing price (Eq. (8)) depends on the demand elasticity v . Therefore, it is imperative that the posterior mean accurately represents the elasticity of demand. Posterior convergence can be assessed by tracking the posterior mean v_{mean} and the HDI width computed from the samples $\{v^{(s)}\}$ at each iteration. As the algorithm converges, we expect the HDI width to decrease and the mean v_{mean} shift so that it is centered on the true elasticity $v = 1.8$.

For reference, a 95% interval corresponds approximately to ± 2 standard deviations under a Normal approximation. In Figure 7, the HDI width decreases to approximately 0.22 at $i = 30$ and $v_{mean} \approx 1.802$. Substituting this v_{mean} into Eq. (8) yields, $p^* = \$2.69$, which is less than 1% error compared to the optimal price.

5.3 Price and Profit KDE density over time (iteration)

KDE (Kernel Density Estimation) plots at each iteration provide additional insight into posterior convergence and uncertainty reduction.

In Figure 8, the KDE of the price distribution, computed from the posterior samples $\{v^{(s)}\}$ based on Eq. (8), is overlaid at each iteration. During the early steps (red), the density is broad and not centered on the optimum price. As the model advances, the distribution becomes increasingly concentrated around the optimal price.

Similarly, the median profit KDE is computed at each step based on the median profit formulation, Eq. (7a). Similar to the price KDE, the profit distribution is relatively broad during the early iterations. As the model advances, the peak of the distribution shifts towards the optimum (median) profit, \$997, and becomes increasingly concentrated.

6. Summary and Conclusions

This paper addresses the need for a cross-functional, rigorous overview of Thompson Sampling for dynamic pricing, including economic stochastic demand modeling, Bayesian inference, and software implementation. The stochastic demand is formulated in detail and clearly shows the demand as a log-normal random variable with multiplicative

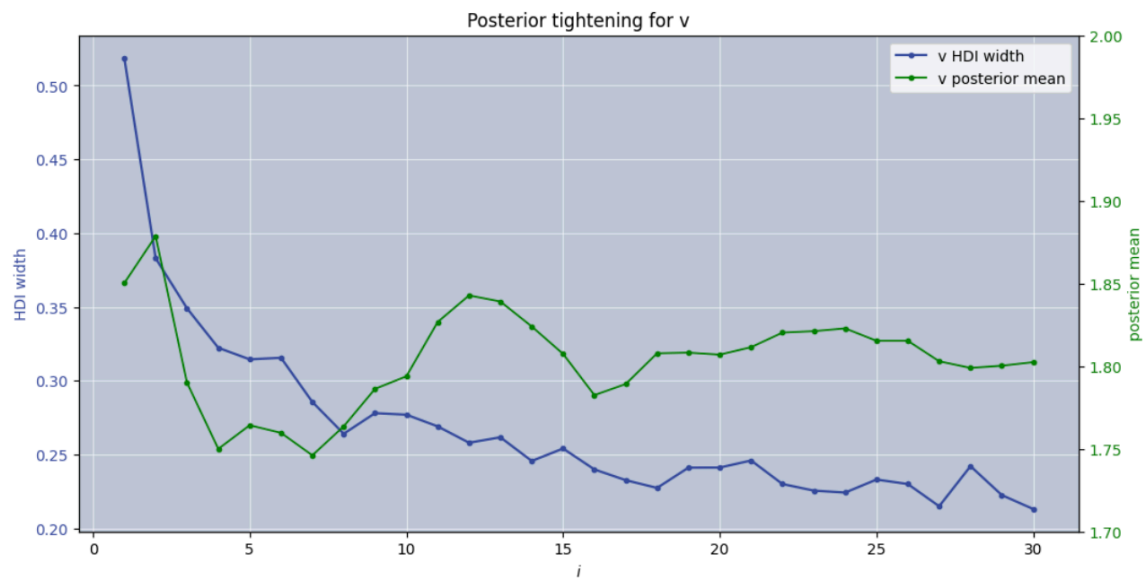
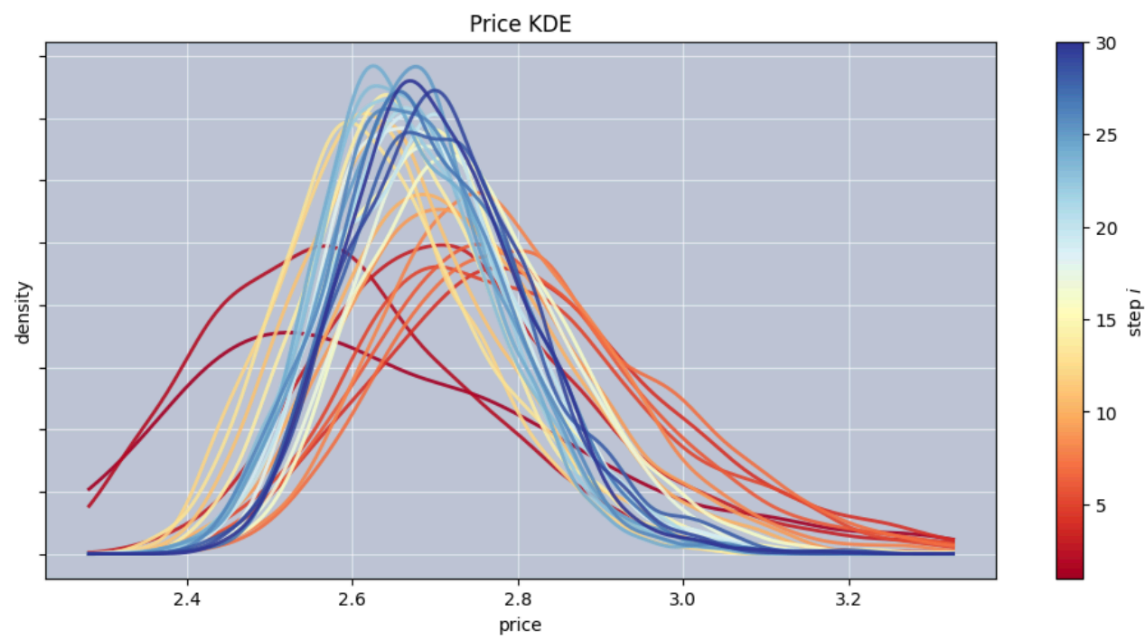
Figure 7: Demand elasticity, v , HDI tightening and mean.

Figure 8: Price KDE at each step.

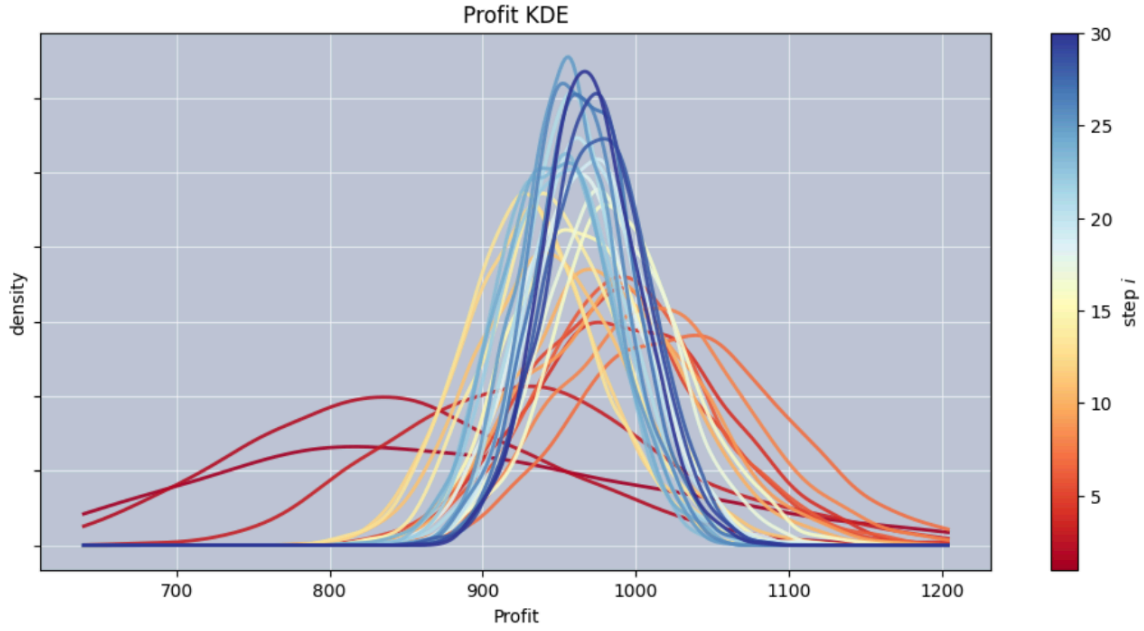


Figure 9: Profit KDE at each step.

demand variation. The resulting stochastic model and software implementation, including diagnostic visualization tools, form a framework for applying TS dynamic pricing. An open-source implementation, based on the PyMC statistical programming library including graphing functions, is available in the accompanying GitHub repository.

Following the development of the stochastic demand model, a key result is the distinction between classical deterministic demand and stochastic demand. Under multiplicative log-normal demand variation, the median demand is equivalent to deterministic demand, while the mean stochastic demand is greater than the median by the factor $e^{\sigma^2/2}$. Thus, accounting for demand shocks results in expected demand, revenue, and profit that exceed their corresponding median values. However, under the constant-elasticity model, the profit-maximizing price remains invariant to multiplicative variations in demand.

The Bayesian posterior statistical pricing model is designed to learn the parameters of our constant elasticity demand model: the multiplicative constant a and the demand elasticity v . By considering the nature of the demand variables, the pricing model is formulated with three statistical components: a log-normal demand likelihood, a log-normal prior a , and a truncated-normal prior v .

A primary insight for achieving accurate convergence is that, under the constant-elasticity model, the optimal price estimate depends solely on the elasticity parameter v (Eq. (8)). Thus, inaccuracies in estimating v directly translate into price inaccuracies. To minimize confounding influences on v , this implementation uses an initial estimate of the demand spread parameter σ (equivalently, CV_d) and recommends a tightly specified prior for a .

The TS pricing algorithm implementation goes beyond the standard (“vanilla”) application of Thompson Sampling. In particular, modeling log-normal demand shocks motivates additional exploration enhancements to improve convergence, including K-posterior sampling, fixed initial prices, repeated prices, and a local price anchor.

Several diagnostic visualizations are used to understand the algorithm’s convergence characteristics. These include price and profit visualized over the iteration steps, showing convergence toward near-optimal values. HDI (Highest Density Interval) visualizations illustrate the step-by-step contraction of posterior densities, indicating that the algorithm learns in the early phases and progressively converges in later phases. In particular, the tightening of the v posterior density can be clearly visualized. KDE overlay plots show the evolution of posterior densities for price and profit: early iterations exhibit broad distributions, while later iterations become increasingly concentrated around the optimum.

Overall, the TS pricing framework demonstrates how sequential Bayesian learning progressively reduces uncertainty in demand estimation and improves pricing decisions over time.

Several areas remain for further study. This work focused on a single product or service; extending the framework to multiple goods, incorporating network or inventory constraints, and retaining realistic stochastic demand models are natural directions for future work. Additional study of realistic demand examples, as well as comparisons with traditional regression-based pricing algorithms, would also be useful.

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